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NEET · CHEMISTRY · CHAPTER-WISE PYQS

Chemical Kinetics

NEET Chemistry · Previous Year Questions with Answer Key

83	44	1994 – 2026	English
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MCQ (Single Correct Answer)	82 questions
MCQ (More than One Correct Answer)	1 question
Answer key	at the end

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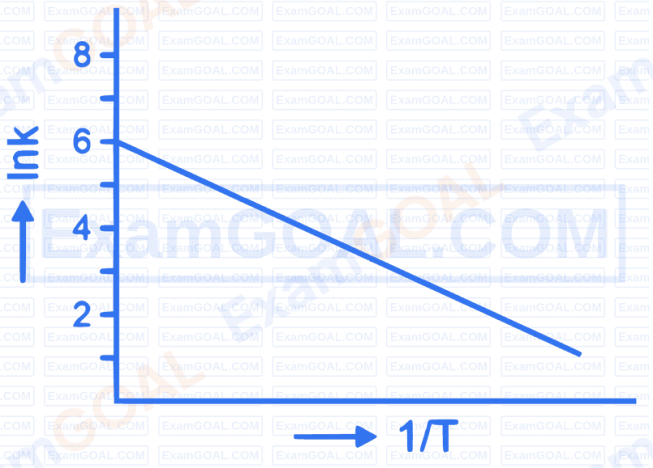
1. **Solve**
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MCQ (Single Correct Answer) 82 questions

Q1. RE-NEET 2026

Open solution

For an elementary chemical reaction, the Arrhenius plot is given below.



If the energy of activation is $6.64\ \text{kJ mol}^{-1}$ and $\text{R} = 8.3\ \text{J K}^{-1}\ \text{mol}^{-1}$, the temperature at which the rate constant becomes $\text{e}^2\ \text{min}^{-1}$, is

(A) 250 K (B) 125 K (C) 150 K (D) 200 K

Q2. RE-NEET 2026

Open solution

$2A \xrightarrow{k} B$ is a zero-order reaction, where $k = 1.0\ \text{mol L}^{-1}\ \text{min}^{-1}$. If the initial concentration of A is 2 M , then the time taken to complete 75% of the reaction will be

(A) 2.0 min (B) 1.5 min (C) 0.75 min (D) 1.0 min

Q3. NEET 2026

Open solution

Match List I with List II :

List-I (Order of reaction)		List-II (Unit of rate constant)	
A.	Zero order	I.	$\text{mol}^{-1}\ \text{L}\ \text{s}^{-1}$
B.	First order	II.	$\text{mol}^{-2}\ \text{L}^2\ \text{s}^{-1}$
C.	Second order	III.	s^{-1}
D.	Third order	IV.	$\text{molL}^{-1}\ \text{s}^{-1}$

Choose the correct answer from the options given below :

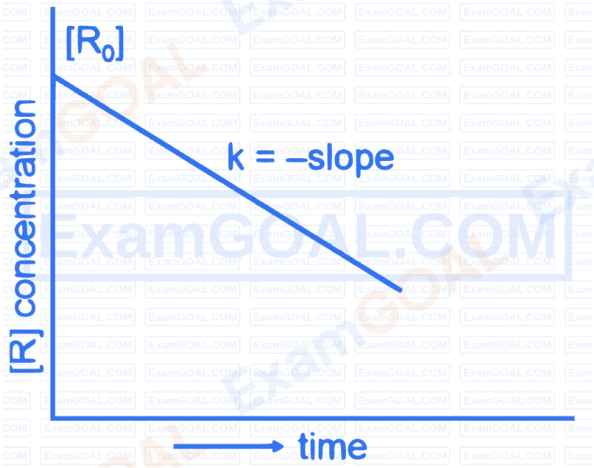
(A) A-IV, B-II, C-I, D-III (B) A-IV, B-III, C-I, D-II

(C) A-IV, B-III, C-II, D-I (D) A-I, B-II, C-III, D-IV

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Q4. NEET 2026

For a certain reaction $R \rightarrow$ Product, the plot of concentration $[R]$ vs time has a negative slope as shown. The order of reaction is :



Open solution

(A) 0 (B) 1 (C) 2 (D) 2.5

Q5. NEET 2026

Given below is an expression for the rate constant of a first-order reaction occurring at a certain temperature, $\mathrm{T(K)}$.

$\ln k = 14.34 - \frac{1.25 \times 10^4}{T}$

The energy of activation in $\mathrm{kcal mol^{-1}}$ for the reaction is :

(Given: $\mathrm{k^{-1}\, s\, s^{-1}}$, $\mathrm{R = 1.987 cal mol^{-1}\, K^{-1}}$)

Open solution

(A) 24.84 (B) 14.34 (C) 18.63 (D) 12.42

Q6. NEET 2025

If the rate constant of a reaction is $0.03\ \mathrm{s^{-1}}$, how much time does it take for $7.2\ \mathrm{mol\ L^{-1}}$ concentration of the reactant to get reduced to $0.9\ \mathrm{mol\ L^{-1}}$? (Given: $\log 2 = 0.301$)

Open solution

(A) 210 s (B) 21.0 s (C) 69.3 s (D) 23.1 s

Q7. NEET 2025

If the half-life ($t_{1/2}$) for a first order reaction is 1 minute, then the time required for 99.9% completion of the reaction is closest to :

Open solution

(A) 5 minutes (B) 10 minutes (C) 2 minutes (D) 4 minutes

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Q8. NEET 2024 (Re-Examination)

Open solution

Following data is for a reaction between reactants A and B :

Rate $\mathrm{mol\,L^{-1}\,s^{-1}}$	$[\mathrm{A}]$	$[\mathrm{B}]$
2×10^{-3}	0.1 M	0.1 M
4×10^{-3}	0.2 M	0.1 M
1.6×10^{-2}	0.2 M	0.2 M

The order of the reaction with respect to A and B, respectively, are

(A) 1, 0 (B) 0, 1 (C) 1, 2 (D) 2, 1

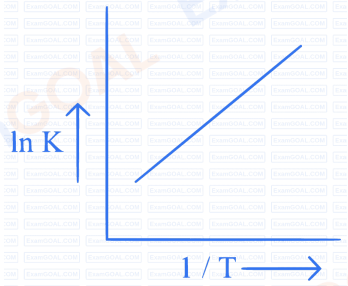
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Q9. NEET 2024 (Re-Examination)

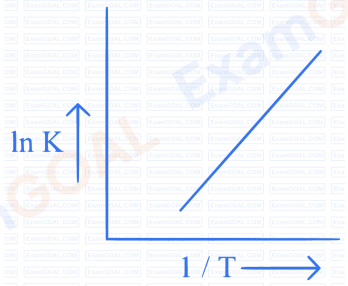
Which of the following plot represents the variation of $\ln k$ versus $\frac{1}{T}$ in accordance with Arrhenius equation?

Open solution

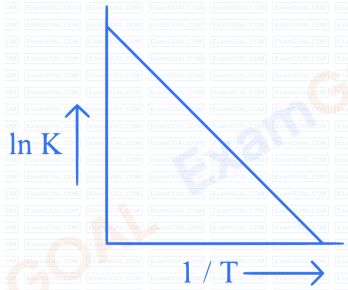
(A)



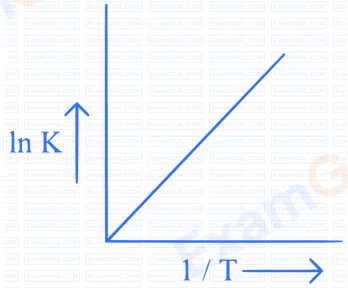
(B)



(C)



(D)



Q10. NEET 2024 (Re-Examination)

Rate constants of a reaction at $500\ \mathrm{K}$ and $700\ \mathrm{K}$ are $0.04\ \mathrm{s}^{-1}$ and $0.14\ \mathrm{s}^{-1}$, respectively; then, activation energy of the reaction is :

(Given: $\log 3.5 = 0.5441, \mathrm{R} = 8.31\ \mathrm{J\ K^{-1}\ mol^{-1}}$)

Open solution

(A) 182310 J (B) 18500 J (C) 18219 J (D) 18030 J

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Q11. NEET 2024

Activation energy of any chemical reaction can be calculated if one knows the value of

(A) rate constant at standard temperature

(B) probability of collision

(C) orientation of reactant molecules during collision

(D) rate constant at two different temperatures

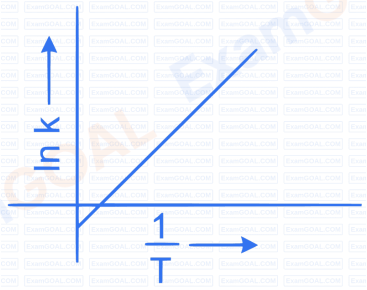
Open solution

Q12. NEET 2024

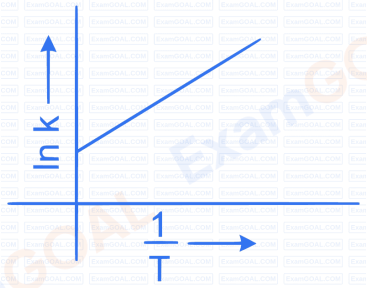
Which plot of $\ln k$ vs $\frac{1}{T}$ is consistent with Arrhenius equation?

Open solution

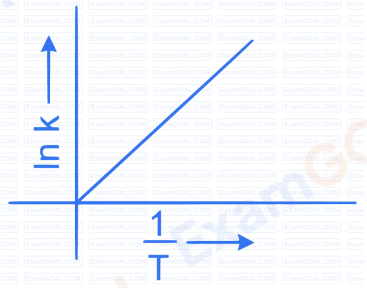
(A)



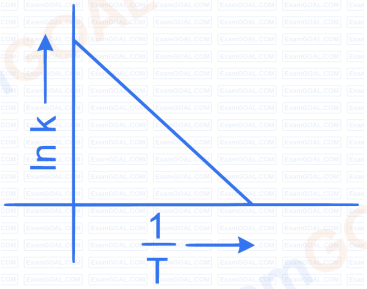
(B)



(C)



(D)



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Q13. NEET 2024

The rate of a reaction quadruples when temperature changes from $27^{\circ}\mathrm{C}$ to $57^{\circ}\mathrm{C}$. Calculate the energy of activation.

Given $\mathrm{R} = 8.314\ \mathrm{J\ K^{-1}\ mol^{-1}}, \log 4 = 0.6021$

Open solution

(A) $38.04\ \mathrm{kJ/mol}$
(B) $380.4\ \mathrm{kJ/mol}$
(C) $3.80\ \mathrm{kJ/mol}$
(D) $3804\ \mathrm{kJ/mol}$

Q14. NEET 2023 Manipur

For a reaction $3\ \mathrm{A} \rightarrow 2\ \mathrm{B}$

The average rate of appearance of B is given by $\frac{\Delta[B]}{\Delta t}$. The correct relation between the average rate of appearance of B with the average rate of disappearance of A is given in option :

Open solution

(A) $\frac{-\Delta[\mathrm{A}]}{\Delta \mathrm{t}}$
(B) $\frac{-3\Delta[\mathrm{A}]}{2\Delta t}$
(C) $\frac{-2\Delta[\mathrm{A}]}{3\Delta \mathrm{t}}$
(D) $\frac{\Delta[\mathrm{A}]}{\Delta t}$

Q15. NEET 2023 Manipur

The correct option for the rate law that corresponds to overall first order reaction is

Open solution

(A) Rate $= \mathrm{k}[\mathrm{A}]^{0}[\mathrm{B}]^{2}$
(B) Rate $= \mathrm{k}[\mathrm{A}][\mathrm{B}]$
(C) Rate $= \mathrm{k}[\mathrm{A}]^{1/2}[\mathrm{B}]^{2}$
(D) Rate $= \mathrm{k}[\mathrm{A}]^{-1/2}[\mathrm{B}]^{3/2}$

Q16. NEET 2023

For a certain reaction, the rate $= \mathrm{k}[\mathrm{A}]^{2}[\mathrm{B}]$, when the initial concentration of A is tripled keeping concentration of B constant, the initial rate would

Open solution

(A) increase by a factor of six
(B) increase by a factor of nine
(C) increase by a factor of three
(D) decrease by a factor of nine

Q17. NEET 2023

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Open solution

Assertion A : A reaction can have zero activation energy.

Reason R : The minimum extra amount of energy absorbed by reactant molecules so that their energy becomes equal to threshold value, is called activation energy.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both A and R are true but R is NOT the correct explanation of A.
(B) A is true but R is false.
(C) A is false but R is true.
(D) Both A and R are true and R is the correct explanation of A.

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Q18. NEET 2022 Phase 2

Open solution

For a chemical reaction

$4A + 3B \rightarrow 6C + 9D$

Rate of formation of C is 6×10^{-2} mol L^{-1} s^{-1} and rate of disappearance of A is 4×10^{-2} mol L^{-1} s^{-1}. The rate of reaction and amount of B consumed in interval of 10 seconds, respectively will be :

(A) 10×10^{-2} mol L^{-1} s^{-1} and 30×10^{-2} mol L^{-1}

(B) 1×10^{-2} mol L^{-1} s^{-1} and 30×10^{-2} mol L^{-1}

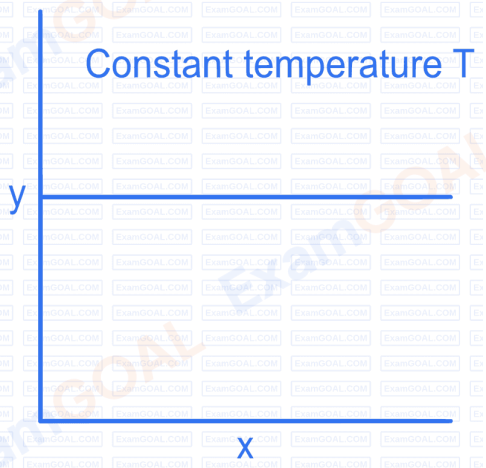
(C) 10×10^{-2} mol L^{-1} s^{-1} and 10×10^{-2} mol L^{-1}

(D) 1×10^{-2} mol L^{-1} s^{-1} and 10×10^{-2} mol L^{-1}

Q19. NEET 2022 Phase 1

Open solution

The given graph is a representation of kinetics of a reaction



The y and x axes for zero and first order reactions, respectively are

(A) zero order (y = concentration and x = time), first order (y = $t_{1/2}$ and x = concentration)

(B) zero order (y = concentration and x = time), first order (y = rate constant and x = concentration)

(C) zero order (y = rate and x = concentration), first order (y = $t_{1/2}$ and x = concentration)

(D) zero order (y = rate and x = concentration), first order (y = rate and x = $t_{1/2}$)

Q20. NEET 2022 Phase 1

Open solution

For a first order reaction A $\rightarrow$ Products, initial concentration of A is 0.1 M, which becomes 0.001 M after 5 minutes. Rate constant for the reaction in min^{-1} is

(A) 1.3818 (B) 0.9212 (C) 0.4606 (D) 0.2303

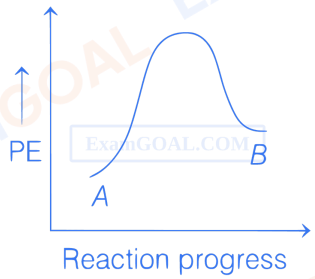
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Q21. NEET 2021

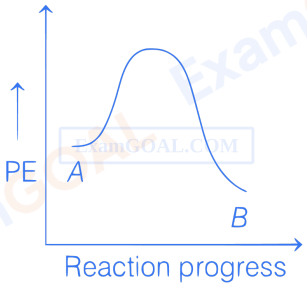
For a reaction, A → B, enthalpy of reaction is -4.2 kJ mol^{-1} and enthalpy of activation is 9.6 kJ mol^{-1}. The correct potential energy profile for the reaction is shown in option.

Open solution

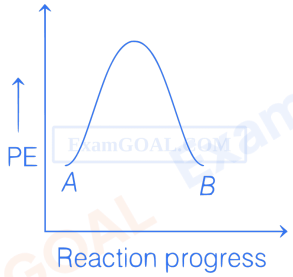
(A)



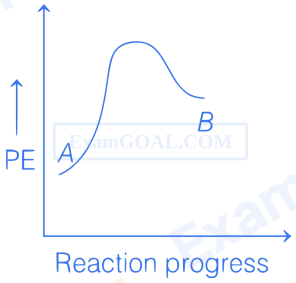
(B)



(C)



(D)



Q22. NEET 2021

The slope of Arrhenius plot $\left(\ln kv/s\frac{1}{T}\right)$ of first order reaction is -5×10^3K. The value of E_a of the reaction is. Choose the correct option for your answer.

Open solution

[Given R = 8.314 JK^{-1}mol^{-1}]

(A) -83 kJ mol^{-1} (B) 41.5 kJ mol^{-1} (C) 83.0 kJ mol^{-1} (D) 166 kJ mol^{-1}

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Q23. NEET 2020 Phase 2

The half-life for a zero order reaction having 0.02 M initial concentration of reactant is 100 s. The rate constant (in $\mathrm{mol\,L^{-1}\,s^{-1}}$) for the reaction is

(A) 1.0×10^{-2} (B) 1.0×10^{-4} (C) 2.0×10^{-4} (D) 2.0×10^{-3}

Open solution

Q24. NEET 2020 Phase 2

In collision theory of chemical reaction, Z_{AB} represents

(A) the fraction of molecules with energies equal to E_{a}

(B) the fraction of molecules with energies greater than E_{a}

(C) the collision frequency of reactants, A and B

(D) steric factor

Open solution

Q25. NEET 2020 Phase 1

An increase in the concentration of the reactants of a reaction leads to change in :

(A) heat of reaction (B) threshold energy

(C) collision frequency (D) activation energy

Open solution

Q26. NEET 2020 Phase 1

The rate constant for a first order reaction is $4.606 \times 10^{-3}\ \mathrm{s^{-1}}$. The time required to reduce 2.0 g of the reactant to 0.2 g is :

(A) 200 s (B) 500 s (C) 1000 s (D) 100 s

Open solution

Q27. NEET 2019 (Odisha)

A first order reaction has a rate constant of $2.303 \times 10^{-3}\ \mathrm{s^{-1}}$. The time required for 40 g of this reactant to reduce to 10 g will be

[Given that $\log_{10} 2 = 0.3010$]

(A) 602 s (B) 230.3 s (C) 301 s (D) 2000 s

Open solution

Q28. NEET 2019 (Odisha)

For a reaction, activation energy $E_a = 0$ and the rate constant at 200 K is $1.6 \times 10^{6}\ \mathrm{s^{-1}}$. The rate constant at 400 K will be

[Given that gas constant, $R = 8.314\ \mathrm{J\,K^{-1}\,mol^{-1}}$]

(A) $3.2 \times 10^{6}\ \mathrm{s^{-1}}$ (B) $3.2 \times 10^{4}\ \mathrm{s^{-1}}$

(C) $1.6 \times 10^{6}\ \mathrm{s^{-1}}$ (D) $1.6 \times 10^{3}\ \mathrm{s^{-1}}$

Open solution

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Q29. NEET 2019

If the rate constant for a first order reaction is k, the time (t) required for the completion of 99 % of the reaction is given by :

(A) $t = \frac{4.606}{k}$ (B) $t = \frac{2.303}{k}$ (C) $t = \frac{0.693}{k}$ (D) $t = \frac{6.909}{k}$

Open solution

Q30. NEET 2019

For the chemical reaction
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
the correct option is :

(A) $-\frac{d[N_2]}{dt} = 2\frac{d[NH_3]}{dt}$
(B) $-\frac{d[N_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$
(C) $3\frac{d[H_2]}{dt} = 2\frac{d[NH_3]}{dt}$
(D) $-\frac{1}{3}\frac{d[H_2]}{dt} = -\frac{1}{2}\frac{d[NH_3]}{dt}$

Open solution

Q31. NEET 2018

The correct difference between first and second order reactions is that

(A) the rate of a first-order reaction does not depend on reactant concentrations; the rate of a second-order reaction does depend on reactant concentrations
(B) the half-life of a first-order reaction does not depend on $[A]_0$; the half-life of a second order reaction does depend on $[A]_0$
(C) a first-order reaction can be catalysed; a second-order reaction cannot be catalysed
(D) the rate of a first-order reaction does depend on reactant concentrations; the rate of a second-order reaction does not depend on reactant concentrations

Open solution

Q32. NEET 2018

When initial concentration of the reactant is doubled, the half-life period of a zero order reaction

(A) is halved (B) is doubled
(C) is tripled (D) remains unchanged.

Open solution

Q33. NEET 2017

Mechanism of a hypothetical reaction
$X_2 + Y_2 \rightarrow 2XY$, is given below :
(i) $X_2 \rightleftharpoons X + X$ (fast)
(ii) $X + Y_2 \rightarrow XY + Y$ (slow)
(iii) $X + Y \rightarrow XY$ (fast)
The overall order of the reaction will be

(A) 2 (B) 0 (C) 1.5 (D) 1

Open solution

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Q34. NEET 2017

A first order reaction has a specific reaction rate of 10^{-2} sec^{-1}. How much time will it take for 20 g of the reactant to reduce to 5 g?

(A) 138.6 sec (B) 346.5 sec (C) 693.0 sec (D) 238.6 sec

Open solution

Q35. NEET 2016 Phase 2

The decomposition of phosphine (PH_3) on tungsten at low pressure is a first-order reaction. It is because the

(A) rate is proportional to the surface coverage

(B) rate is inversely proportional to the surface coverage

(C) rate is independent of the surface coverage

(D) rate of decomposition is very slow.

Open solution

Q36. NEET 2016 Phase 1

The rate of first-order reaction is 0.04 mol L^{-1} s^{-1} at 10 seconds and 0.03 mol L^{-1} s^{-1} at 20 seconds after initiation of the reaction. The half-life period of the reaction is

(A) 44.1 s (B) 54.1 s (C) 24.1 s (D) 34.1 s

Open solution

Q37. NEET 2016 Phase 1

The addition of a catalyst during a chemical reaction alters which of the following quantities?

(A) Enthalpy (B) Activation energy

(C) Entropy (D) Internal energy

Open solution

Q38. AIPMT 2015

The rate constant of the reaction A $\rightarrow$ B is 0.6×10^{-3} mol L^{-1} s^{-1}. If the concentration of A is 5 M, then concentration of B after 20 minutes is

(A) 3.60 M (B) 0.36 M (C) 0.72 M (D) 1.08 M

Open solution

Q39. AIPMT 2015 Cancelled Paper

The activation energy of a reaction can be determined from the slope of which of the following graphs?

(A) ln k vs. $\frac{1}{T}$ (B) $\frac{T}{\ln k}$ vs. $\frac{1}{T}$

(C) ln k vs. T (D) $\frac{\ln k}{T}$ vs. T

Open solution

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Q40. AIPMT 2015 Cancelled Paper

When initial concentration of a reactant is doubled in a reaction, its half-life period is not affected. The order of the reaction is

(A) second
(B) more than zero but less than first
(C) zero
(D) first.

Open solution

Q41. NEET 2013 (Karnataka)

For a reaction between A and B the order with respect to A is 2 and the other with respect to B is 3. The concentrations of both A and B are doubled, the rate will increase by a factor of

(A) 12
(B) 16
(C) 32
(D) 10

Open solution

Q42. NEET 2013 (Karnataka)

A reaction is 50% complete in 2 hours and 75% complete in 4 hours. The order of reaction is

(A) 1
(B) 2
(C) 3
(D) 0

Open solution

Q43. NEET 2013

What is the activation energy for a reaction if its rate doubles when the temperature is raised from $20^{o}C$ to $35^{o}C$?
($R = 8.314\ J\ mol^{-1}\ K^{-1}$)

(A) $34.7\ kJ\ mol^{-1}$
(B) $15.1\ kJ\ mol^{-1}$
(C) $342\ kJ\ mol^{-1}$
(D) $269\ kJ\ mol^{-1}$

Open solution

Q44. AIPMT 2012 Prelims

In a reaction, A + B $\rightarrow$ product, rate is doubled when the concentration of B is doubled, and rate increases by a factor of 8 when the concentration of both the reactants (A and B) are doubled, rate law for the reaction can be written as

(A) rate = $k[A][B]^2$
(B) rate = $k[A]^2[B]^2$
(C) rate = $k[A][B]$
(D) rate = $k[A]^2[B]$

Open solution

Q45. AIPMT 2012 Prelims

In a zero-order reaction, for every $10^{o}C$ rise of temperature, the rate is doubled. If the temperature is increased from $10^{o}C$ to $100^{o}C$, the rate of the reaction will become

(A) 256 times
(B) 512 times
(C) 64 times
(D) 128 times

Open solution

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Q46. AIPMT 2011 Mains

The rate of the reaction : $2N_2O_5 \rightarrow 4NO_2 + O_2$ can be written in three ways.

$\frac{-d[N_2O_5]}{dt} = k\,[N_2O_5]$

$\frac{d[NO_2]}{dt} = k'\,[N_2O_5]$; $\frac{d[O_2]}{dt} = k''\,[N_2O_5]$

The relationship between k and k' and between k and k" are

(A) $k' = 2k; k'' = k$ (B) $k' = 2k; k'' = k/2$

(C) $k' = 2k; k'' = 2k$ (D) $k' = k; k'' = k$

Open solution

Q47. AIPMT 2011 Mains

The unit of rate constant for a zero order reaction is

(A) mol L^{-1} s^{-1} (B) L mol^{-1} s^{-1} (C) L^2 mol^{-2} s^{-1} (D) s^{-1}

Open solution

Q48. AIPMT 2011 Mains

The half-life of a substance in a certain enzyme-catalysed reaction is 138 s. The time required for the concentration of the substance to fall from 1.28 mg L^{-1} to 0.04 mg L^{-1} is

(A) 414 s (B) 552 s (C) 690 s (D) 276 s

Open solution

Q49. AIPMT 2011 Prelims

Which one of the following statements for the order of a reaction is incorrect?

(A) Order can be determined only experimentally.

(B) Order is not influenced by stoichiometric coefficient of the reactants.

(C) Order of a reaction is sum of power to the concentration terms of reactants to express the rate of reaction.

(D) Order of reaction is always whole number.

Open solution

Q50. AIPMT 2010 Mains

The rate of the reaction, $2NO + Cl_2 \rightarrow 2NOCl$ is given by the rate equation rate = $k[NO]^2[Cl_2]$. The value of the rate constant can be increased by

(A) increasing the temperature (B) increasing the concentration of NO

(C) increasing the concentration of the Cl_2 (D) doing all of these.

Open solution

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Q51. AIPMT 2010 Prelims

Open solution

For the reaction $N_2O_{5(g)} \rightarrow 2NO_{2(g)} + 1/2O_{2(g)}$ the value of rate of disappearance of N_2O_5 is given as 6.25×10^{-3} mol L^{-1} s^{-1}. The rate of formation of NO_2 and O_2 is given respectively as

(A) 6.25×10^{-3} mol L^{-1} s^{-1} and 6.25×10^{-3} mol L^{-1} s^{-1}

(B) 1.25×10^{-2} mol L^{-1} s^{-1} and 3.125×10^{-3} mol L^{-1} s^{-1}

(C) 6.25×10^{-3} mol L^{-1} s^{-1} and 3.125×10^{-3} mol L^{-1} s^{-1}

(D) 1.25×10^{-2} mol L^{-1} s^{-1} and 6.25×10^{-3} mol L^{-1} s^{-1}

Q52. AIPMT 2010 Prelims

Open solution

During the kinetic study of the reaction, $2A + B \rightarrow C + D$, following results were obtained

Run	[A]/mol L^{-1}	[B]/mol L^{-1}	Initial rate of formation of D/mol L^{-1} min^{-1}
I.	0.1	0.1	6.0×10^{-3}
II.	0.3	0.2	7.2×10^{-2}
III.	0.3	0.4	2.88×10^{-1}
IV.	0.4	0.1	2.40×10^{-2}

Based on the above data which one of the following is correct?

(A) Rate = $k[A]^2[B]$ (B) Rate = $k[A][B]$ (C) Rate = $k[A]^2[B]^2$ (D) Rate = $k[A][B]^2$

Q53. AIPMT 2009

Open solution

For the reaction, $N_2 + 3H_2 \rightarrow 2NH_3$, if $\frac{d[NH_3]}{dt} = 2 \times 10^{-4}$ mol L^{-1} s^{-1}, the value of $\frac{-d[H_2]}{dt}$ would be

(A) 4×10^{-4} mol L^{-1} s^{-1} (B) 6×10^{-4} mol L^{-1} s^{-1}

(C) 1×10^{-4} mol L^{-1} s^{-1} (D) 3×10^{-4} mol L^{-1} s^{-1}

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Q54. AIPMT 2009

Open solution

In the reaction,

$BrO^-_{3(aq)} + 5Br^-_{(aq)} + 6H^+ \rightarrow 3Br_{2(l)} + 3H_2O_{(l)}$.

The rate of appearance of bromine (Br_2) is related to rate of disappearance of bromide ions as

(A) $\frac{d[Br_2]}{dt} = -\frac{5}{3}\frac{d[Br^-]}{dt}$

(B) $\frac{d[Br_2]}{dt} = \frac{5}{3}\frac{d[Br^-]}{dt}$

(C) $\frac{d[Br_2]}{dt} = \frac{3}{5}\frac{d[Br^-]}{dt}$

(D) $\frac{d[Br_2]}{dt} = -\frac{3}{5}\frac{d[Br^-]}{dt}$

Q55. AIPMT 2009

Open solution

Half-life period of a first order reaction is 1386 seconds. The specific rate constant of the reaction is

(A) $0.5 \times 10^{-2}\ s^{-1}$

(B) $0.5 \times 10^{-3}\ s^{-1}$

(C) $5.0 \times 10^{-2}\ s^{-1}$

(D) $5.0 \times 10^{-3}\ s^{-1}$.

Q56. AIPMT 2009

Open solution

For the reaction A + B $\rightarrow$ products, it is observed that

(i) on doubling the initial concentration of A only, the rate of reaction is also doubled and
(ii) on doubling the initial concentration of both A and B, there is a change by a factor of 8 in the rate of the reaction.

The rate of this reaction is given by

(A) rate = $k[A]^2[B]^2$

(B) rate = $k[A][B]^2$

(C) rate = $k[A][B]$

(D) rate = $k[A]^2[B]$

Q57. AIPMT 2008

Open solution

The rate constants k_1 and k_2 for two different reactions are $10^{16} \cdot e^{-2000/T}$ and $10^{15} \cdot e^{-1000/T}$, respectively.

The temperature at which $k_1 = k_2$ is

(A) 2000 K

(B) $\frac{1000}{2.303} K$

(C) 1000 K

(D) $\frac{2000}{2.303} K$

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Q58. AIPMT 2008

Open solution

The bromination of acetone that occurs in acid solution is represented by this equation.

$CH_3COCH_{3(aq)} + Br_{2(aq)} \rightarrow$

$CH_3COCH_2Br(aq) + H^+_{(aq)} + Br^-_{(aq)}$

These kinetic data were obtained for given reaction concentrations.

Initial concentrations, M				
$[CH_3COCH_3]$		$[Br_2]$		$[H^+]$
0.30		0.05		0.05
0.30		0.10		0.05
0.30		0.10		0.10
0.40		0.05		0.20

Initial rate, disappearance of Br_2, Ms^{-1}						
			5.7×10^{-5}			
			5.7×10^{-5}			
			1.2×10^{-4}			
			3.1×10^{-4}			

Based on these data, the rate equation is

(A) Rate = $k[CH_3COCH_3][Br_2][H^+]^2$

(B) Rate = $k[CH_3COCH_3][Br_2][H^+]$

(C) Rate = $k[CH_3COCH_3][H^+]$

(D) Rate = $k[CH_3COCH_3][Br_2]$

Q59. AIPMT 2007

Open solution

If 60% of a first order reaction was completed in 60 minutes, 50% of the same reaction would be completed in approximately
(log 4 = 0.60, log 5 = 0.69)

(A) 45 minutes (B) 60 minutes (C) 40 minutes (D) 50 minutes

Q60. AIPMT 2007

Open solution

In a first-order reaction A $\rightarrow$ B, if k is rate constant and initial concentration of the reactant A is 0.5 M, then the half-life is

(A) $\frac{\log 2}{k}$ (B) $\frac{\log 2}{k\sqrt{0.5}}$ (C) $\frac{\ln 2}{k}$ (D) $\frac{0.693}{0.5k}$

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Q61. AIPMT 2007

The reaction of hydrogen and iodine monochloride is given as :
$H_{2(g)} + 2ICl_{(g)} \rightarrow 2HCl_{(g)} + I_{2(g)}$
This reaction is of first order with respect to $H_{2(g)}$ and $ICl_{(g)}$, following mechanisms were proposed.

Mechanism A :
$H_{2(g)} + 2ICl_{(g)} \rightarrow 2HCl_{(g)} + I_{2(g)}$
Mechanism B :
$H_{2(g)} + ICl_{(g)} \rightarrow HCl_{(g)} + HI_{(g)}$; slow
$HI_{(g)} + ICl_{(g)} \rightarrow HCl_{(g)} + I_{2(g)}$; fast

Which of the above mechanism(s) can be consistent with the given information about the reaction?

(A) *A* and *B* both (B) Neither *A* nor *B* (C) *A* only (D) *B* only

Open solution

Q62. AIPMT 2006

Consider the reaction : $N_{2(g)} + 3H_{2(g)} \rightarrow 2NH_{3(g)}$

The equality relationship between $\frac{d[NH_3]}{dt}$ and $-\frac{d[H_2]}{dt}$ is

(A) $\frac{d[NH_3]}{dt} = -\frac{d[H_2]}{dt}$
(B) $\frac{d[NH_3]}{dt} = -\frac{1}{3}\frac{d[H_2]}{dt}$
(C) $+\frac{d[NH_3]}{dt} = -\frac{2}{3}\frac{d[H_2]}{dt}$
(D) $+\frac{d[NH_3]}{dt} = -\frac{3}{2}\frac{d[H_2]}{dt}$

Open solution

Q63. AIPMT 2006

For the reaction, $2A + B \rightarrow 3C + D$, which of the following does not express the reaction rate?

(A) $-\frac{d[A]}{2dt}$ (B) $-\frac{d[C]}{3dt}$
(C) $-\frac{d[B]}{dt}$ (D) $\frac{d[D]}{dt}$

Open solution

Q64. AIPMT 2005

The rate of reaction between two reactants A and B decreases by a factor of 4 if the concentration of reactant B is doubled. The order of this reaction with respect to reactant B is

(A) 2 (B) -2 (C) 1 (D) -1

Open solution

Q65. AIPMT 2005

For a first order reaction $A \rightarrow B$ the reaction rate a reactant concentration of 0.01 M is found to be 2.0×10^{-5} mol L^{-1} s^{-1}. The half-life period of the reaction is

(A) 30 s (B) 220 s (C) 300 s (D) 347 s

Open solution

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Q66. AIPMT 2004

The rate of a first order reaction is 1.5×10^{-2} mol L^{-1} min^{-1} at 0.5 M concentration of the reactant. The half-life of the reaction is

(A) 0.383 min (B) 23.1 min (C) 8.73 min (D) 7.53 min

Open solution

Q67. AIPMT 2003

The temperature dependence of rate constant (k) of a chemical reaction is written in terms of Arrhenius equation, $k = A \cdot e^{-E^*/RT}$. Activation energy (E^*) of the reaction can be calculated by plotting

(A) $k \; vs \; T$ (B) $k \; vs \; \frac{1}{\log T}$

(C) $\log \; k \; vs \; \frac{1}{T}$ (D) $\log \; k \; vs \; \frac{1}{\log T}$

Open solution

Q68. AIPMT 2003

If the rate of the reaction is equal to the rate constant, the order of the reaction is

(A) 0 (B) 1 (C) 2 (D) 3

Open solution

Q69. AIPMT 2003

The reaction A $\rightarrow$ B follows first order kinetics. The time taken for 0.8 mole of A to produce 0.6 mole of B is 1 hour. What is the time taken for conversion of 0.9 mole of A to produce 0.675 mole of B?

(A) 1 hour (B) 0.5 hour (C) 0.25 hour (D) 2 hours

Open solution

Q70. AIPMT 2003

The activation energy for a simple chemical reaction A $\rightleftharpoons$ B is E_a in forward direction.

The activation energy for reverse reaction

(A) is negative of E_a (B) is always less than E_a

(C) can be less than or more than E_a (D) is always double of E_a

Open solution

Q71. AIPMT 2002

2A $\rightarrow$ B + C

It would be a zero order reaction when

(A) the rate of reaction is proportional to square of concentration of A

(B) The rate of reaction remains same at any concentration of A

(C) the rate remains unchanged at any concentration of B and C

(D) the rate of reaction doubles if concentrations of B is increased to double.

Open solution

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Q72. AIPMT 2002

$3A \rightarrow 2B$, rate of reaction $+\frac{d[B]}{dt}$ is equal to

(A) $-\frac{3}{2}\frac{d[A]}{dt}$ (B) $-\frac{2}{3}\frac{d[A]}{dt}$ (C) $-\frac{1}{3}\frac{d[A]}{dt}$ (D) $+2\frac{d[A]}{dt}$

Open solution

Q73. AIPMT 2001

When a bio-chemical reaction is carried out in laboratory, outside the human body in absence of enzyme, then rate of reaction obtained is 10^{-6} times, the activation energy of reaction in the presence of enzyme is

(A) 6/RT
(B) P is required
(C) different from E_a obtained in laboratory
(D) can't say anything.

Open solution

Q74. AIPMT 2001

For the reaction;

$2N_2O_5 \rightarrow 4NO_2 + O_2$ rate and rate constant are 1.02×10^{-4} and 3.4×10^{-5} sec^{-1} respectively, then concentration of N_2O_5 at that time will be

(A) 1.732 (B) 3 (C) 1.02×10^{-4} (D) 3.4×10^{5}

Open solution

Q75. AIPMT 2000

For the reaction $H^+ + BrO_3^- + 3Br^- \rightarrow 5Br_2 + H_2O$

which of the following relation correctly represents the consumption and formation of products.

(A) $\frac{d[Br^-]}{dt} = -\frac{3}{5}\frac{d[Br_2]}{dt}$
(B) $\frac{d[Br^-]}{dt} = \frac{3}{5}\frac{d[Br_2]}{dt}$
(C) $\frac{d[Br^-]}{dt} = -\frac{5}{3}\frac{d[Br_2]}{dt}$
(D) $\frac{d[Br^-]}{dt} = \frac{5}{3}\frac{d[Br_2]}{dt}$

Open solution

Q76. AIPMT 1999

For a first-order reaction, the half-life period is independent of

(A) initial concentration
(B) cube root of initial concentration
(C) first power of final concentration
(D) square root of final concentration

Open solution

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Q77. AIPMT 1998

Activation energy of a chemical reaction can be determined by

(A) evaluating rate constant at standard temperature

(B) evaluating velocities of reaction at two different temperatures

(C) evaluating rate constants at two different temperatures

(D) changing concentration of reactants

Open solution

Q78. AIPMT 1997

The experimental data for the reaction, $2A + B_2 \rightarrow 2AB$ is

Experiment	$[A]$	$[B_2]$	Rate (mole s^{-1})
1	0.50	0.50	1.6×10^{-4}
2	0.50	1.00	3.2×10^{-4}
3	1.00	1.00	3.2×10^{-4}

The rate equation for the above data is

(A) Rate $= k[A]^2[B]^2$ (B) Rate $= k[A]^2[B]$ (C) Rate $= k[B_2]$ (D) Rate $= k[B_2]^2$

Open solution

Q79. AIPMT 1997

For the reaction $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$, the rate of reaction is expressed as

(A) $\frac{\Delta[H_2]}{\Delta t} = \frac{1}{2}\frac{\Delta[I_2]}{\Delta t} = -\frac{\Delta[HI]}{\Delta t}$

(B) $-\frac{\Delta[I_2]}{\Delta t} = -\frac{\Delta[H_2]}{\Delta t} = \frac{1}{2}\frac{\Delta[HI]}{\Delta t}$

(C) $\frac{\Delta[I_2]}{\Delta t} = \frac{\Delta[H_2]}{\Delta t} = \frac{\Delta[HI]}{2\Delta t}$

(D) none of these

Open solution

Q80. AIPMT 1996

The given reaction,

$2FeCl_3 + SnCl_2 \rightarrow 2FeCl_2 + SnCl_4$

is an example of

(A) third order reaction (B) first order reaction

(C) second order reaction (D) none of these

Open solution

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Q81. AIPMT 1995

A chemical reaction is catalyzed by a catalyst *X*. Hence *X*

(A) reduces enthalpy of the reaction

(B) does not affect equilibrium constant of reaction

(C) decreases rate constant of the reaction

(D) increases activation energy of the reaction.

Open solution

Q82. AIPMT 1994

The data for the reaction $A + B \rightarrow C$, is

Exp.	$[A]_0$	$[B]_0$	Initial rate
1	0.012	0.035	0.10
2	0.024	0.070	0.80
3	0.024	0.035	0.10
4	0.012	0.070	0.80

The rate law corresponds to the above data is

(A) Rate $= k[A][B]^3$ (B) Rate $= k[A]^2[B]^2$ (C) Rate $= k[B]^3$ (D) Rate $= k[B]^4$

Open solution

MCQ (More than One Correct Answer) 1 questions

Q83. AIPMT 2012 Mains

Activation energy (E_a) and rate constants (k_1 and k_2) of a chemical reaction at two different temperatures (T_1 and T_2) are related by

(A) $\ln \frac{k_2}{k_1} = -\frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$

(B) $\ln \frac{k_2}{k_1} = -\frac{E_a}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$

(C) $\ln \frac{k_2}{k_1} = -\frac{E_a}{R}\left(\frac{1}{T_2} + \frac{1}{T_1}\right)$

(D) $\ln \frac{k_2}{k_1} = \frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **D**	2. **C**	3. **B**	4. **A**	5. **A**
6. **C**	7. **B**	8. **C**	9. **C**	10. **C**
11. **D**	12. **D**	13. **A**	14. **C**	15. **D**
16. **B**	17. **C**	18. **B**	19. **C**	20. **B**
21. **B**	22. **B**	23. **B**	24. **C**	25. **C**
26. **B**	27. **A**	28. **C**	29. **A**	30. **B**
31. **B**	32. **B**	33. **C**	34. **A**	35. **A**
36. **C**	37. **B**	38. **C**	39. **A**	40. **D**
41. **C**	42. **A**	43. **A**	44. **D**	45. **B**
46. **B**	47. **A**	48. **C**	49. **D**	50. **A**
51. **B**	52. **D**	53. **D**	54. **D**	55. **B**
56. **B**	57. **B**	58. **C**	59. **A**	60. **C**
61. **D**	62. **C**	63. **B**	64. **B**	65. **D**
66. **B**	67. **C**	68. **A**	69. **A**	70. **C**
71. **B**	72. **B**	73. **C**	74. **B**	75. **A**
76. **A**	77. **C**	78. **C**	79. **B**	80. **A**
81. **B**	82. **C**			

MCQ (More than One Correct Answer)

83. **B, D**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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